

Practice Problems:
Inductance and Capacitance
Introduction to Electric Circuits
Supplemental Instruction

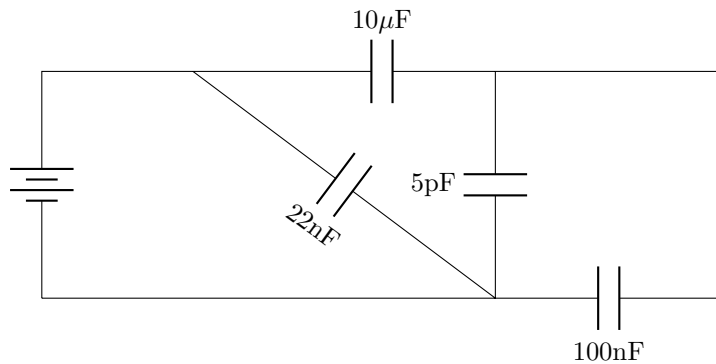
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This document is not a substitute for a textbook or classroom instruction. It may contain errors and is intended to be used only to the extent that it is helpful to supplement students' learning experience.

1 Capacitor Combinations

Find the total capacitance as seen from the source.

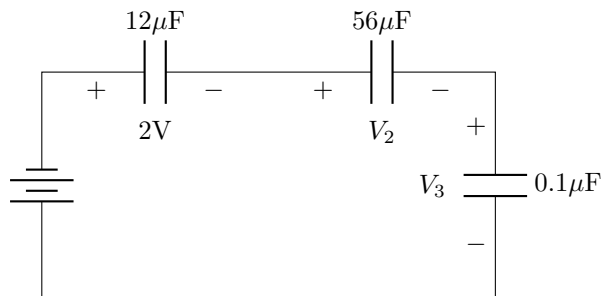


Solution

The two capacitors on the right can be combined in parallel: $500\text{pF} + 100\text{nF} = 100.5\text{nF}$. This can then be combined in series with the capacitor on the top: $\frac{1}{\frac{1}{100.5\text{nF}} + \frac{1}{10\mu\text{F}}} = 99.5\text{nF}$. Finally, combining in parallel with the 22nF capacitor: $99.5\text{nF} + 22\text{nF} = 121.5\text{nF}$.

Capacitor Charge Given Voltage

Find the total charge stored by the capacitors, V_2 , and V_3 .

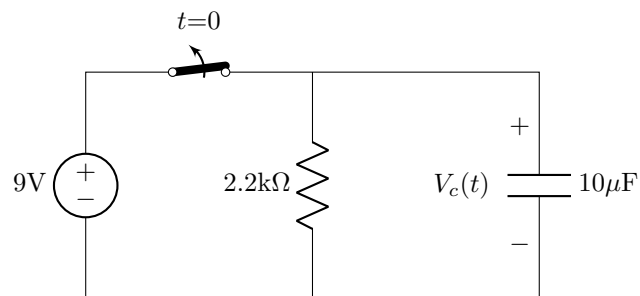


Solution

The charge on the leftmost capacitor can be found as $Q = CV = 12\mu\text{F} \cdot 2\text{V} = 24\mu\text{C}$. Since the charge is the same across series capacitors, we can find immediately $V_2 = \frac{24\mu\text{C}}{56\mu\text{F}} = 428\text{mV}$, and $V_3 = \frac{24\mu\text{C}}{0.1\mu\text{F}} = 240\text{V}$.

2 RC Circuit Response

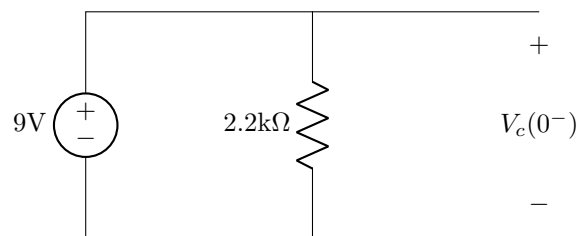
The switch is closed for a long time and then it is opened at $t = 0$. Find (a) the equation of the voltage across the capacitor after the switch opens (b) the instantaneous voltage across the capacitor at $t = 5\text{ms}$, and (c) the time that it will take for the capacitor to be fully discharged.



Solution

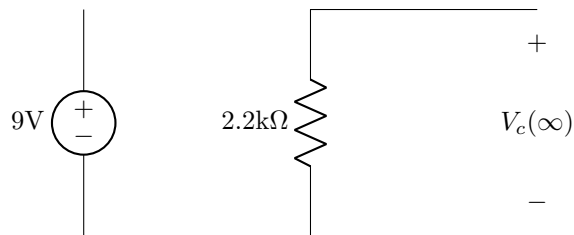
The time constant can be found as $\tau = RC = 2.2\text{k}\Omega \cdot 10\mu\text{F} = 22\text{ms}$.

Before the switch is opened, at $t = 0^-$, the capacitor is an open circuit:



$V_c(0^-) = 9\text{V}$. Since the voltage of a capacitor cannot change instantly, $V_c(0^+) = V_c(0^-) = 9\text{V}$.

After the switch has been open for a long time, the capacitor is again an open circuit:



$$V_c(\infty) = 0\text{V}.$$

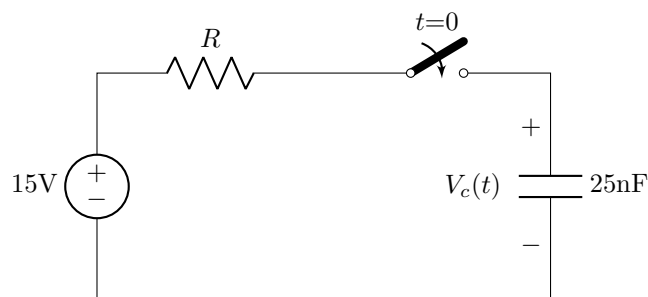
Therefore, the voltage across the capacitor over time can be found as $V_c(t) = (V_c(0^+) - V_c(\infty))e^{\frac{-t}{\tau}} + V_c(\infty) \Rightarrow V_c(t) = (9\text{V} - 0\text{V})e^{\frac{-t}{22\text{ms}}} + 0\text{V} = 9 \cdot e^{\frac{-t}{22\text{ms}}} \text{V}$ (part a).

At $t = 5\text{ms}$, the voltage is $V_c(5\text{ms}) = 9 \cdot e^{\frac{-5\text{ms}}{22\text{ms}}} \text{V} = 7.17 \text{V}$ (part b).

The capacitor will be fully discharged after $5\tau = 5 \cdot 22\text{ms} = 110\text{ms}$ (part c).

3 RC Circuit Response

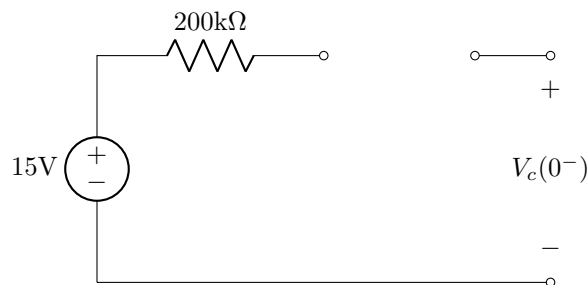
Find the value of R so that the circuit will have a time constant of $\tau = 5\text{ms}$, and find the equation of $V_c(t)$ for $t > 0$ using that value of R .



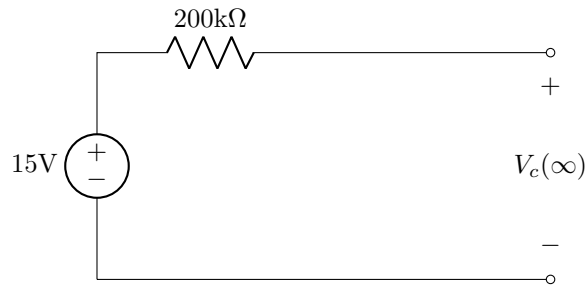
Solution

R can be found as $\frac{\tau}{C} = \frac{5\text{ms}}{25\text{nF}} = 200\text{k}\Omega$.

Next, we'll redraw the circuit at $t = 0^-$, and replace the capacitor with an open circuit:



The capacitor is not connected on the top, so $V_c(0^-) = V_c(0^+) = 0\text{V}$. Next, we'll redraw the circuit at $t = \infty$, again at steady state:



There is current flowing through the resistor, so $V_c(\infty) = 15\text{V}$.

Now we can substitute the values we found into the first order response equation:

$$V_c(t) = (V_c(0^+) - V_c(\infty))e^{\frac{-t}{\tau}} + V_c(\infty) \Rightarrow V_c(t) = [(0 - 15)e^{\frac{-t}{5\text{ms}}} + 15] \text{ V} = [-15e^{\frac{-t}{5\text{ms}}} + 15] \text{ V}.$$

References

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- M. Iordache. 2024. EEGR 2053 Electric circuits [PowerPoint slides]. Available: <https://mviordache.name/EEGR2053/index.html>
- T. L. Floyd and D. M. Buchla, Principles of Electric Circuits: Conventional Current, 10th ed. Hoboken, NJ: Pearson Education, 2020.
- O. Ortiz. 2025. ENGR 2053 Intro to electric circuits [PowerPoint slides].